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LiteFNDpp: A Dual-Path Lightweight and Interpretable Model for Real-Time Fake News Detection

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Software

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Summary

The rapid spread of online misinformation poses significant challenges to public trust and information integrity. Recent approaches to fake news detection increasingly rely on transformer-based language models, which achieve strong predictive performance but incur substantial computational costs and offer limited interpretability (Brown & others, 2020; Devlin et al., 2019; Liu & others, 2019). These constraints restrict their applicability in real-time and resource-constrained deployment scenarios.

LiteFNDpp is a lightweight and interpretable dual-path ensemble framework designed for efficient fake news detection on CPU-only systems. The model combines term frequency–inverse document frequency (TF–IDF) representations with a calibrated soft-voting ensemble of Logistic Regression and Multinomial Naive Bayes classifiers (Joachims, 1998; Rennie & others, 2003). Linguistically informed preprocessing, including semantic normalization and named-entity-aware token handling (Bird et al., 2009; Honnibal & Montani, 2017), improves robustness to stylistic and adversarial variations in news content.

To promote transparency, LiteFNDpp integrates Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016), enabling human-readable explanations for individual predictions. Empirical evaluation demonstrates that the system achieves competitive classification performance while maintaining low inference latency, making it suitable for real-world misinformation monitoring applications where efficiency and interpretability are essential.

Statement of Need

Most state-of-the-art fake news detection systems employ deep neural networks or transformer-based architectures. Although effective, these models typically require specialized hardware, introduce inference latency, and provide limited insight into the rationale behind predictions (Lipton, 2018). Such limitations hinder adoption in domains where explainability, responsiveness, and accessibility are critical.

LiteFNDpp addresses this gap by providing an open-source software package that emphasizes computational efficiency, interpretability, and ease of deployment. The framework is particularly valuable for researchers, journalists, and developers who require a transparent and real-time fake news detection solution without reliance on GPUs or large-scale infrastructure. By combining classical machine learning techniques with modern explainability tools, LiteFNDpp offers a practical alternative to heavyweight deep learning approaches.

Related Work

LiteFNDpp builds on prior interpretability-focused software such as `FAT Forensics` and ensemble-based NLP classifiers (Castillo et al., 2011; Ruchansky et al., 2017; Shu et al.,

2017). Unlike these tools, LiteFNDpp focuses on **real-time fake news detection** with a **dual-path lightweight ensemble** that achieves CPU-efficient inference with integrated LIME-based explanations.

Implementation

LiteFNDpp is implemented in Python and leverages widely used NLP libraries (Bird et al., 2009; Honnibal & Montani, 2017) and machine learning tools. Text preprocessing includes:

- Lowercasing, punctuation removal, and token filtering
- Named-entity-aware transformations (e.g., "Barack Obama" → "Barack_Obama")
- Semantic normalization of dates, numerals, and locations

Feature representations are constructed using **word-level and character-level TF–IDF vectors** (Manning et al., 2008; Zhang et al., 2015) to capture lexical and subword patterns.

The classification pipeline consists of a **dual-path ensemble**:

- **Logistic Regression (LR)** – handles high-dimensional TF–IDF features (Joachims, 1998)

- **Multinomial Naive Bayes (NB)** – models term-frequency distributions (McCallum & Nigam, 1998; Rennie & others, 2003)

Predictions are combined via **weighted soft voting**:

$$P_{\text{final}}(y) = 0.65 \cdot P_{\text{LR}}(y) + 0.35 \cdot P_{\text{NB}}(y)$$

Explanations are generated using **LIME**, producing localized and human-readable interpretations (Ribeiro et al., 2016).

The software architecture is **modular and extensible**, allowing users to modify preprocessing, classifiers, or explanation mechanisms.

Usage Example

```python
from litefndpp.model import LiteFNDpp

# Initialize the model
model = LiteFNDpp()

# Train the model
X_train = ["News article text 1", "News article text 2"]
y_train = [1, 0]  # 1 = Fake, 0 = Real
model.fit(X_train, y_train)

# Predict
prediction = model.predict(["Breaking: miracle cure discovered!"])

# Explain prediction
explanation = model.explain("Breaking: miracle cure discovered!")
print(prediction, explanation)
```

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Features

- **Dual-Path Ensemble:** Logistic Regression + Naive Bayes via weighted soft voting
- **CPU-Efficient:** ~16 MB model size, inference ~12 ms
- **Linguistically Informed Preprocessing:** NER-aware token joining, semantic normalization
- **TF–IDF Feature Engineering:** Word n-grams (1–4) and character n-grams (3–6)
- **Explainability:** LIME provides token-level insights into predictions
- **Modular Architecture:** Customizable preprocessing, models, and explainability tools

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Performance

Model	Accu-racy	F1-Score	Inference Time (ms)	Explainable
BERT-base (Devlin et al., 2019)	97.8	0.979	243	No
RoBERTa-base (Liu & others, 2019)	98.2	0.981	221	No
DistilBERT (Sanh et al., 2019)	97.6	0.976	112	No
TF–IDF + LR	98.4	0.983	17	Yes
LiteFNDpp	99.0	0.991	12	Yes

- LiteFNDpp demonstrates competitive classification performance while substantially reducing inference latency on CPU-only systems.
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Ablation studies indicate that the dual-path ensemble, named-entity-aware token joining, and semantic normalization each contribute to overall performance.

Evaluation

- **Cross-Dataset Generalization:** Trained on FakeNewsNet (Shu et al., 2020) and evaluated on the LIAR dataset, demonstrating consistent performance across datasets.
- **Explainability Analysis:** LIME provides token-level explanations that highlight words contributing to *Fake* or *Real* predictions (Ribeiro et al., 2016).
- **Computational Efficiency:** The trained model occupies approximately 16 MB of memory, and training completes in approximately five minutes on standard CPU hardware.

Ablation Study Example (F1-score)

Configuration	F1-Score
TF–IDF + LR (baseline)	0.983
+ Naive Bayes Ensemble	0.987
+ NER-aware Token Joining	0.989
+ Semantic Normalization (Full LiteFNDpp)	0.991

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